

Trainings ▾

Preparatory Steps

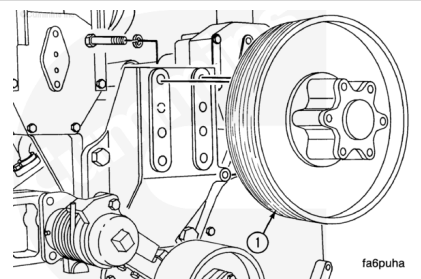
- Remove the fan. Refer to Procedure 008-040 in Section 8. (</qs3/pubsys2/xml/en/procedures/57/57-008-040-tr.html>)
- Remove the fan belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/57/57-008-002-tr.html>)



Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



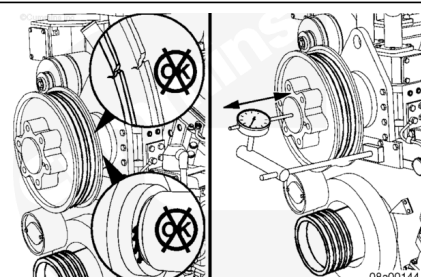
Remove the four capscrews, washers, and the fan hub assembly from the fan support.

Remove the fan hub assembly (1).

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturers recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

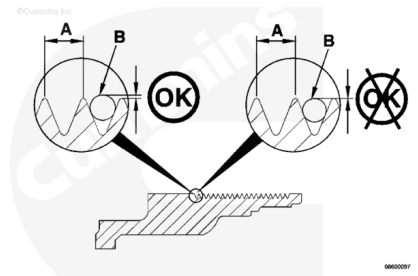
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Rotate the fan hub pulley to check for rough or damaged bearings. Inspect the pulley grooves for excessive wear. Check for grease leakage.

If damage to the pulley is found, it **must** be replaced. If grease leakage is found, the fan hub **must** be rebuilt.

Poly Vee pulley groove wear can be measured by placing a gauge ball or drill bit in the pulley groove and checking the protrusion with a straight edge. If the gauge ball or drill bit is below the pulley peaks, the pulley **must** be replaced.

To determine which gauge ball or drill bit can be used, measure the A dimension and compare to the dimensions of K and L section pulleys.

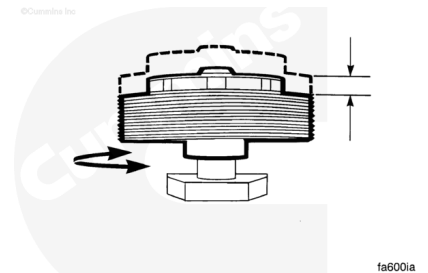


Poly Vee - Pulley Reuse Guidelines

Belt Type	A	B
K-Section	3.56 mm [0.140 in]	3 mm [0.12 in] gauge ball or a number 31 drill bit
L-Section	4.70 mm [0.185 in]	4 mm [0.16 in] gauge ball or a number 20 drill bit

Use a dial indicator to check the bearing end clearance.

Bearing End Clearance		
mm		in
0.025	MIN	0.001
0.152	MAX	0.006



If the clearance is **not** within the specifications, the fan hub **must** be rebuilt.

Rotate the pulley to check for rough or damaged bearings

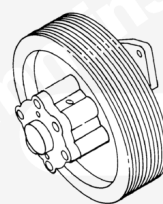
If the fan hub does **not** rotate freely, the fan hub **must** be rebuilt.

Use solvent. Clean the fan hub.

Dry with compressed air.

Torque Value: 15 n•m [133 in-lb]

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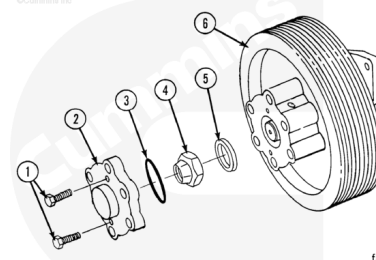
Disassemble

Remove the following parts from the pulley (6).

1. Capscrews
2. Fan pilot
3. O-ring
4. Locknut
5. Plain hardened washer.

Discard the o-ring.

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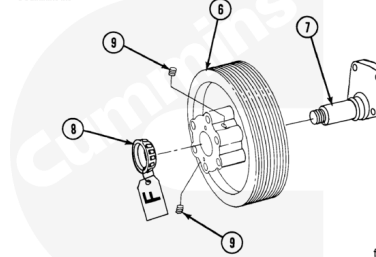
Note : If force is required, use an arbor press. Support the pulley (6) and push on the small end of the shaft (7).

Remove the pulley assembly (6) from the shaft (7).

Remove and tag the front roller bearing (8) for future identification.

Remove the pipe plugs (9), if fitted.

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Turn the pulley assembly (6) over.

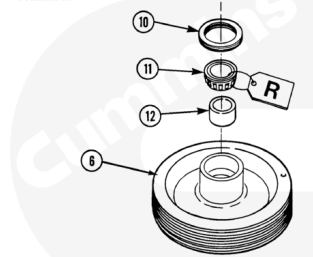
Remove the oil seal (10).

Remove the rear roller bearing (11) and tag for future identification.

Remove the rear bearing spacer (12).

Note : Do not remove the front bearing spacer.

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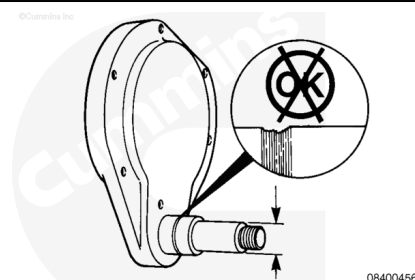
Assemble

Check for wear at the seal location.

Measure the shaft at the bearing location.

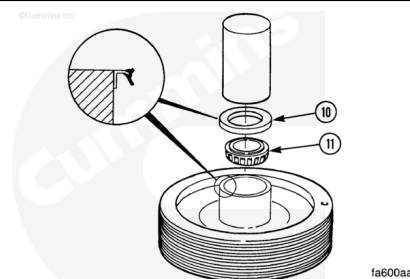
Shaft Outside Diameter		
mm		in
49.91	MIN	1.965
49.98	MAX	1.968

If the shaft is out of specification or excessive wear is seen, the shaft **must** be replaced.



⚠ CAUTION ⚠

Do not lubricate the seal surface of the shaft. The seal and seal surface must be clean and dry. Failure to do so can result in seal failure.



Note : Excessive grease will affect the end float reading.

Lightly grease the bearing (11).

Use high performance, general purpose industrial-type grease (NLGI Grade No. 2) Chevron™ SRI Grease 2 or Mobilux™ EP 2 grease, or its equivalent.

If above greases are **not** available locally, contact a Cummins® Distributor for advice on other Cummins® approved equivalents.

Install the bearing in the hub.

Apply Loctite® 675, or equivalent, to the outside diameter of the seal.

Install the seal (10). Use a mandrel with the flat unseamed face toward the shoulder of the driver and press the seal in until it bottoms.

Install the fan hub seal end first over the shaft and make sure no damage occurs to the seal.

The rear bearing **must** slide easily on the shaft. If it does **not**, check for chips or burrs.

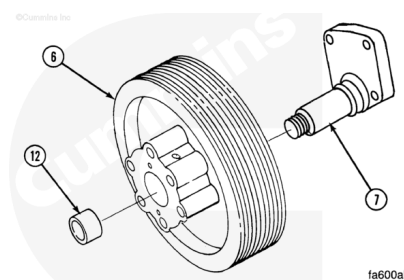
Install the inner bearing spacer (12). The following spacers are available to provide correct end clearance when the fan hub has been assembled.

The following listing is for two spacer style fan hubs:

- 3627681 - 63.906 mm [2.5160 in] length
- 3627682 - 63.995 mm [2.5195 in] length
- 3627683 - 64.084 mm [2.5230 in] length
- 3627684 - 64.173 mm [2.5265 in] length.

The following listing is for other fan hubs:

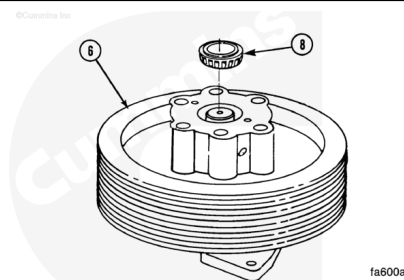
- 3627677 - 51.562 mm [2.0300 in] length
- 3627678 - 51.689 mm [2.0350 in] length
- 3627679 - 51.816 mm [2.0400 in] length
- 3627680 - 51.943 mm [2.0450 in] length
- 3628830 - 52.019 mm [2.0480 in] length.



Lightly grease the front bearing (8).

Install the bearing into the pulley (6).

The bearing **must** slide easily over the shaft. If it does **not**, check for chips and burrs.



⚠ CAUTION ⚠

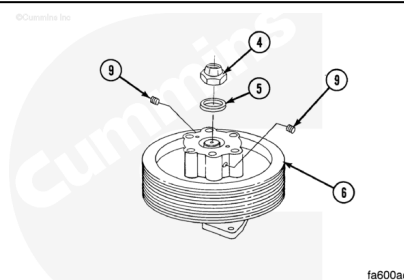
Remove all grease from the threads on the shaft before installing the locknut. Grease on the threads reduce the torque retention of the locknut.

Install the hardened washer (5) into the pulley (6).

Install the lock nut (4).

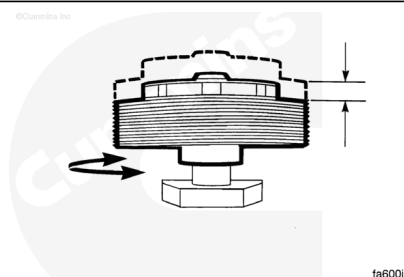
Torque Value: 610 n•m [450 ft-lb]

Install the pipe plug (9).



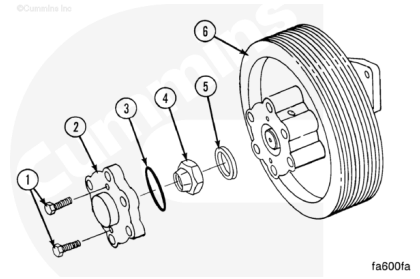
Use a dial indicator to measure the bearing end clearance.

Compare the measurement to the Bearing End Clearance specifications listed above.



⚠ CAUTION ⚠

Do not use more grease than specified. Too much grease will cause excessive heating and failure.



Use high performance, general purpose industrial-type grease (NLGI Grade No. 2) Chevron™ SRI Grease 2 or Mobilux™ EP 2 grease, or its equivalent.

If above greases are **not** available locally, contact a Cummins® Distributor for advice on other Cummins® approved equivalents.

Fill the cavity with 120 cc of grease. Grease the inner race, fit the spacer and inner race.

Install the hardened washer (5). Install the lock nut (4).

Check the bearing end clearance.

Install the o-ring (3) on the fan pilot (2).

Lubricate the o-ring with vegetable oil.

Install the fan pilot and o-ring.

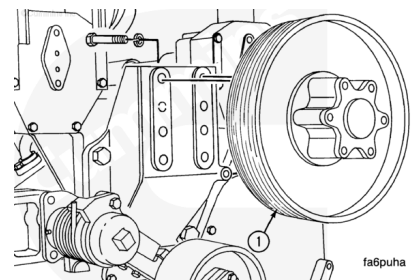
Install the capscrews (1).

Torque Value: 20 n•m [177 in-lb]

Install

⚠ WARNING ⚠

This component weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly..



Install the fan hub assembly (1), and the four capscews and washers, to the fan support.

Tighten the fan hub capscrews.

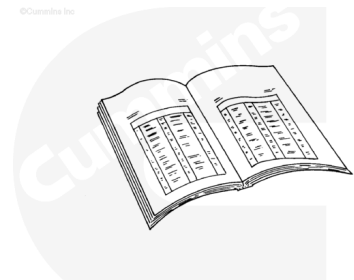
Torque Value: 180 n•m [133 ft-lb]

Note : Go to the fan option for fan hub centerlines and mounting locations.

Finishing Steps

⚠ WARNING ⚠

This component weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



ck800wa

- Install the fan belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/57/57-008-002-tr.html>)
- Install the fan. Refer to Procedure 008-040 in Section 8. (</qs3/pubsys2/xml/en/procedures/57/57-008-040-tr.html>)
- Operate the engine and check for proper operation.